

LANGUAGE: PYTHON PROGRAMMING
LEVEL:INTERMEDIATE-ADVANCED

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	Intermediate–Advanced Corporate Use Case
Day 1	Python Introduction & Environment Setup	Python Overview, Python 2 vs Python 3, Advantages & Disadvantages of Python, Installation, IDE Setup, Python Interpreter, pip, Virtual Environment, pydoc, help(), dir(), UNIT TESTING Basics, PEP8 Coding Standards	LeetCode: Fizz Buzz, Palindrome Number HackerRank: Python If-Else, Python Arithmetic Operators	Automating Manual Excel Reports in Finance Teams (Migration from VBA to Python Reporting Tool)
Day 2	Python Fundamentals	Variables, Keywords, Built-in Functions, Data Types, Numeric Literals, Strings, Operators, Expressions, Type Conversion, Input/Output, String Manipulation	LeetCode: Valid Anagram, Reverse Integer HackerRank: Python Strings, Find a String	Employee Data Validation Engine before SAP/ERP Integration
Day 3	Flow Control & Loops	Conditional Statements, Relational & Boolean Operators, if-else, for/while loops, Nested Loops, Loop Optimization, Pattern Programs	LeetCode: Fizz Buzz, Power of Two HackerRank: Loops, Nested Lists	Automated IT Ticket Routing & Priority Assignment System
Day 4	Functions & Functional Programming	Function Definitions, Parameters, Variable Scope, *args/**kwargs, Lambda Functions, map/filter/reduce, Decorators, Generators, Recursive Functions	LeetCode: Climbing Stairs, Two Sum HackerRank: Python Functions, Map and Lambda Function	Reusable Enterprise Validation Framework used across Multiple Internal Portals
Day 5	Python Collections & Comprehensions	Lists, Tuples, Dictionaries, Sets, Frozen Sets, List/Dict/Set Comprehensions, Nested Collections, Slicing, Packing & Unpacking, enumerate(), zip(), Counter, defaultdict, deque, OrderedDict, Iterators & Iterables	LeetCode: Group Anagrams, Contains Duplicate, Majority Element HackerRank: Lists, Dictionaries and Maps, Nested Lists	Customer Order Processing & Enterprise Metadata Dashboard
Day 6	Object-Oriented Programming (OOP)	Classes, Objects, Constructors, Inheritance, Method Overriding, Encapsulation, Abstraction, Polymorphism, Static & Class Methods, Property Decorators	LeetCode: Design Parking System, Merge Intervals HackerRank:	Enterprise HR Hierarchy & Role Management System

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			Classes and Objects	
Day 7	Exception Handling & Debugging	Standard Exception Hierarchy, try-except-finally, Custom Exceptions, Assertions, Logging, Debugging Techniques, PEP8 Compliance	LeetCode: Missing Number, Single Number HackerRank: Exceptions	Payroll Failure Recovery & Secure Transaction Validation System
Day 8	File Handling & Data Processing	Reading/Writing Files, with Block, CSV Library, JSON Handling, XML Basics, Log File Parsing, Data Serialization	LeetCode: Valid Sudoku, Matrix Diagonal Sum HackerRank: File I/O	Security Audit Log Analysis & Monitoring Tool
Day 9	Modules, Packages & Database Connectivity	Modules, Packages, import Statement, pip Installation, sqlite3, CRUD Operations, Transactions, Database Connectivity Concepts	LeetCode: Rotate Array, Majority Element HackerRank: SQLite Basics	Enterprise Employee Database & Attendance Management System
Day 10	NumPy, Pandas & Data Analytics	NumPy Arrays, Indexing, Broadcasting, Pandas DataFrames, Data Cleaning, Aggregation, Filtering, GroupBy, Merging, ETL Processing	LeetCode: Maximum Subarray, Best Time to Buy & Sell Stock HackerRank: Numpy Arrays, Shape and Reshape	Enterprise Sales & Customer Analytics Dashboard
Day 11	Matplotlib, Data Science & Scikit-Learn	Data Visualization, Line/Bar/Pie Charts, Machine Learning Basics, Classification, Regression, Train-Test Split, Model Evaluation, Feature Engineering	LeetCode: Binary Search, Merge Sorted Array HackerRank: Statistics and Machine Learning Basics	Predictive Employee Attrition & Customer Churn Analysis System
Day 12	Advanced Python & Corporate Problem Solving	Multithreading, Multiprocessing, REST API Basics, JSON APIs, Web Scraping, Automation Scripts, Performance Optimization, Heapq, deque, Advanced Collections Usage	LeetCode: Longest Substring Without Repeating Characters, Top K Frequent Elements, Merge Intervals HackerRank: Python Itertools, Collections	Real-Time Enterprise Data Processing & Automation Pipeline
Day 13	Advanced Coding + RDBMS Concepts	First Half – Advanced Coding: Mock Coding Interviews, Pandas-based Optimization, Collections-Based Coding,	LeetCode: LRU Cache, Valid Parentheses, Contains	Enterprise Data Warehouse & Reporting System

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		Dictionary & Set Optimization, Sliding Window Problems, Debugging & Refactoring. Second Half – RDBMS: DBMS vs RDBMS, ER Model, Keys, Normalization, SQL Queries, Joins, Aggregate Functions, Transactions	Duplicate HackerRank: SQL Basic Select, Weather Observation Station, Dynamic Array	
Day 14	Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Cloud Computing Basics, IaaS/PaaS/SaaS, Git & GitHub Workflow, clone/push/pull/merge conflicts, Docker Basics, Docker Images & Containers, Docker Compose, Kubernetes Basics, K8S Commands, CI/CD Pipelines, GitHub Actions, Jenkins Basics	LeetCode: Single Number, Move Zeroes, Merge Intervals HackerRank: Linux Shell Basics, Git Tutorials, 30 Days of Code	End-to-End Cloud Native Python Microservices Deployment Pipeline using Docker + Kubernetes + CI/CD

Additional Interview-Oriented Python Collections Topics

Collection Topic	Key Concepts	Corporate Use Case
List	append(), extend(), slicing, sorting	Employee Records Processing
Tuple	Immutable Data Handling	Financial Configurations
Dictionary	key-value processing, update(), get()	API Response Mapping
Set	Duplicate Removal, Set Operations	Fraud Detection Pipelines
Counter	Frequency Analysis	Log Monitoring Systems
defaultdict	Missing Key Handling	ETL Data Pipelines
deque	Queue & Streaming Processing	Real-Time Messaging Systems
OrderedDict	Ordered Transactions	Banking Transaction Logs
Heapq	Priority Queue	Task Scheduling Engine
zip() & enumerate()	Batch Iteration	Reporting Automation

Important Corporate Python Libraries

Library	Usage
NumPy	Numerical Computing
Pandas	Data Analysis & ETL
Matplotlib	Data Visualization
Scikit-Learn	Machine Learning
requests	API Integration
json	JSON Processing
sqlite3	Database Connectivity
threading	Multithreading
multiprocessing	Parallel Processing
logging	Enterprise Logging

Recommended Platforms for Practice

- [LeetCode](#)
- [HackerRank Python](#)
- [GeeksforGeeks Python](#)
- [NumPy Documentation](#)
- [Pandas Documentation](#)
- [Scikit-Learn Documentation](#)



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